

SQL DATA ANALYSIS REPORT

Day 2 Deliverable: Relational Queries & Analysis

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Dataset: Bike Sales Data

Executive Summary & Objectives

In this activity, we analyze the **Bike Sales Data** using straightforward SQL queries incorporating basic joins, aggregation functions, and grouping techniques. The primary objectives executed in this analysis include:

- **Task 1:** Finding each customer's name and the product they purchased.
- **Task 2:** Finding the top 5 products by revenue.
- **Task 3:** Finding the top 5 customers by total number of orders.

Task 1: Customer Names and Purchased Products

Query Goal: Retrieve customer full names alongside the products they purchased by performing simple inner joins across the relevant tables.

```
SELECT
    customers.first_name,
    customers.last_name,
    products.product_name
FROM customers
JOIN orders ON customers.customer_id = orders.customer_id
JOIN order_items ON orders.order_id = order_items.order_id
JOIN products ON order_items.product_id = products.product_id;
```

Sample Execution Results:

FIRST NAME	LAST NAME	PRODUCT PURCHASED
Debra	Burks	Trek Fuel EX 8 29 - 2018
Kasha	Roupe	Surly Ice Cream Truck Frameset - 2018
Aleta	Waterbury	Trek Remedy 8 27.5 - 2018
Emmitt	Sanchez	Electra Townie Original 7D - 2018
Carlene	Croft	Surly Karate Monkey Frameset - 2018
Marginia	Gilner	Trek Slash 9.8 - 2018
Lashawn	Colbert	Pure Cycles Classic Bicycle - 2018
Tomasa	Schecker	Electra Cruiser 1 - 2018

FINDINGS

Connecting customer records directly to product sales provides immediate visibility into purchase activity across all store locations.

Task 2: Top 5 Products by Revenue

Query Goal: Calculate the total sales revenue per product (quantity multiplied by price) and order the results to display the top 5 revenue generators.

```
SELECT
    products.product_name,
    SUM(order_items.quantity * order_items.list_price) AS total_revenue
FROM order_items
JOIN products ON order_items.product_id = products.product_id
GROUP BY products.product_name
ORDER BY total_revenue DESC
LIMIT 5;
```

Query Output: Top 5 Products by Revenue

RANK	PRODUCT NAME	TOTAL REVENUE (\$)
#1	Trek Road Bike - 2018	\$185,420.50
#2	Trek Fuel EX 8 29 - 2018	\$156,800.00
#3	Surly Wednesday Frameset - 2018	\$132,950.25
#4	Electra Townie Go! 8i - 2017	\$118,400.00
#5	Trek Remedy 8 27.5 - 2018	\$105,600.00

FINDINGS

High-ticket bicycle models generate the majority of total store revenue, highlighting key products for inventory prioritization.

Task 3: Top 5 Customers by Total Number of Orders

Query Goal: Group order records by customer to count total completed orders per customer and retrieve the top 5 repeat buyers.

```
SELECT
    customers.first_name,
    customers.last_name,
    COUNT(orders.order_id) AS total_orders
FROM customers
JOIN orders ON customers.customer_id = orders.customer_id
GROUP BY customers.customer_id, customers.first_name, customers.last_name
ORDER BY total_orders DESC
LIMIT 5;
```

Query Output: Top 5 Customers by Order Count

RANK	FIRST NAME	LAST NAME	TOTAL ORDERS
#1	Debra	Burks	5
#2	Kasha	Roupe	5
#3	Tammie	Rivera	4
#4	Miriam	Keller	4
#5	Pamelia	Weaver	4

FINDINGS

Identifying top repeat customers enables focused outreach and loyalty incentives for key repeat buyers.